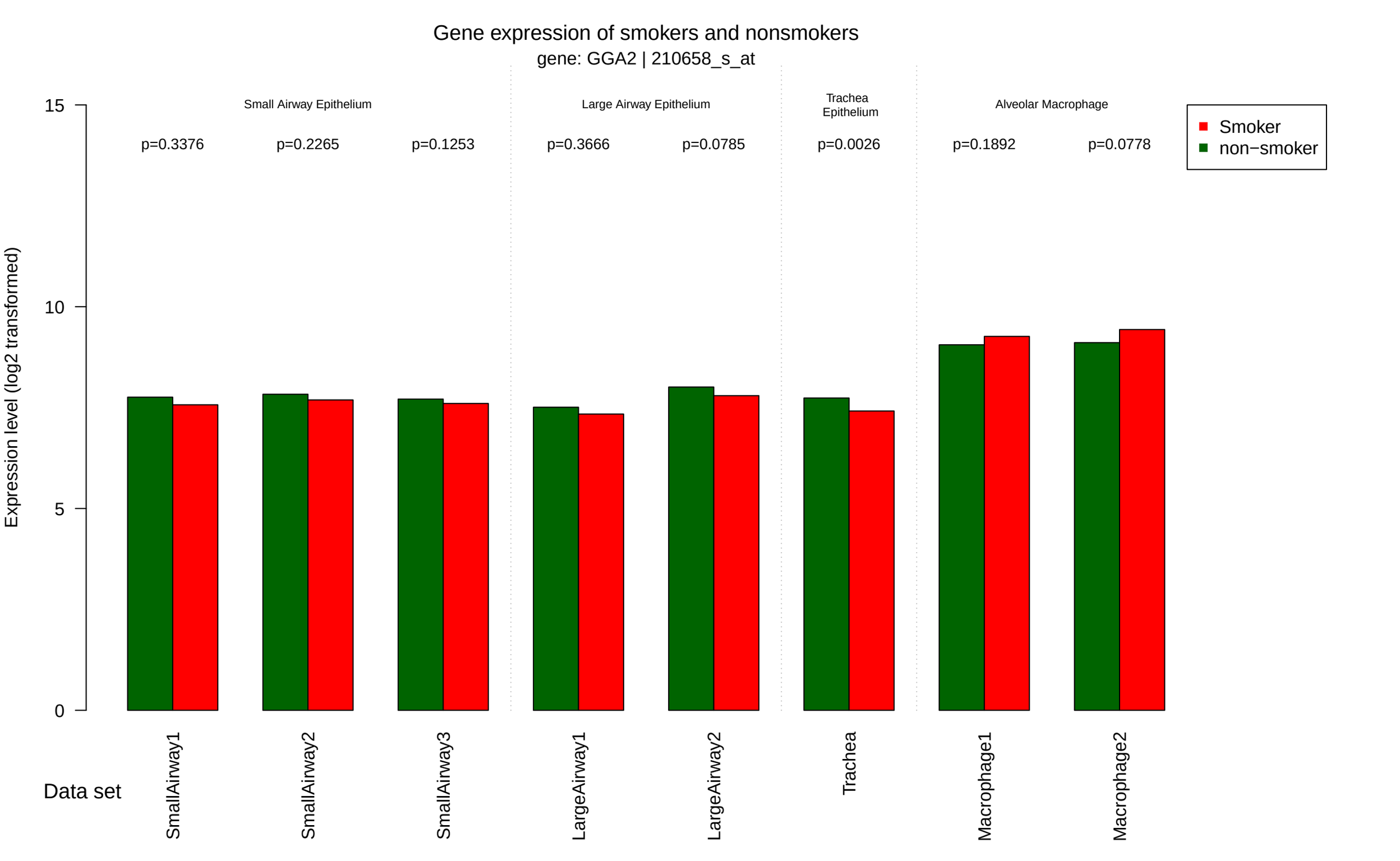
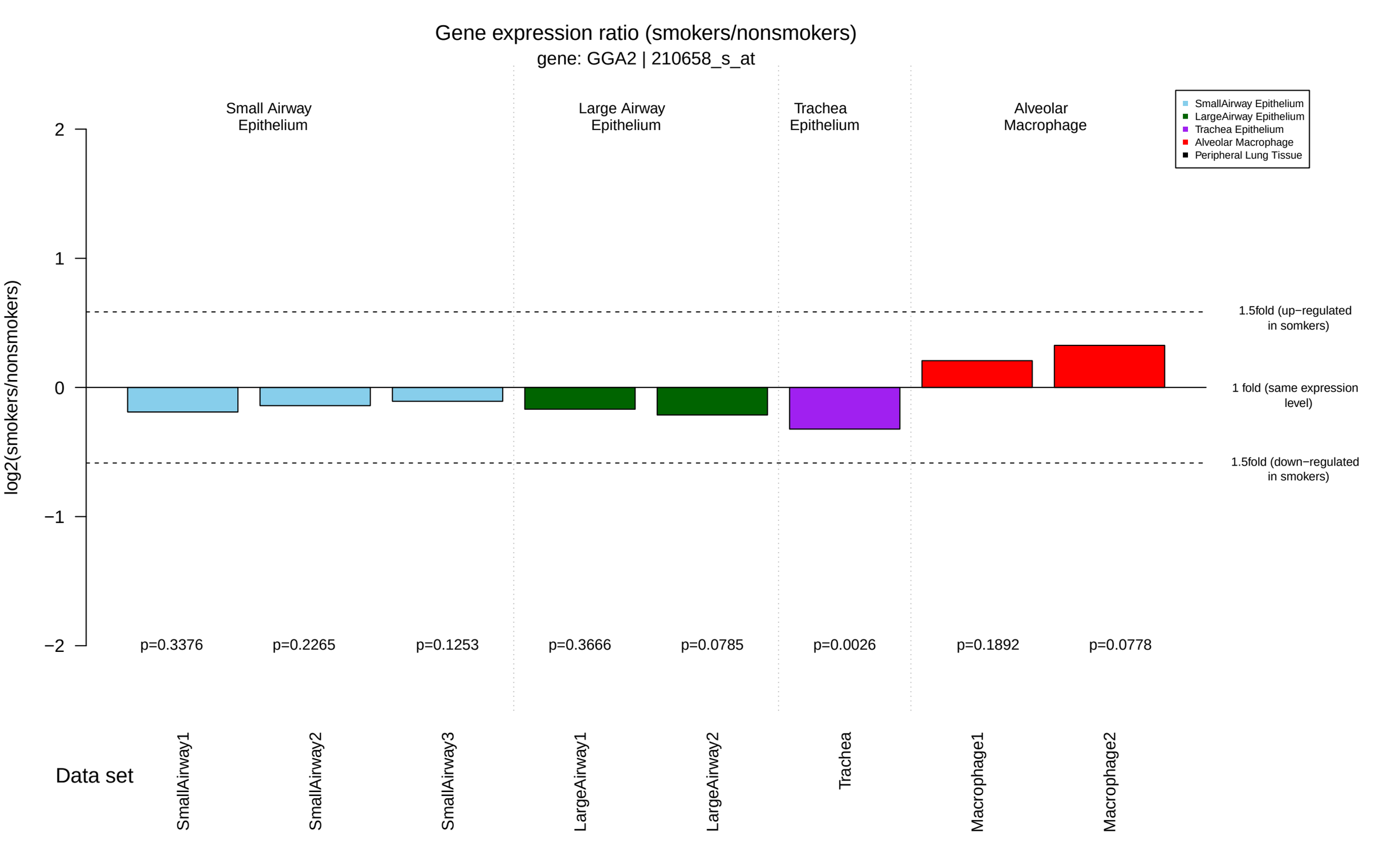
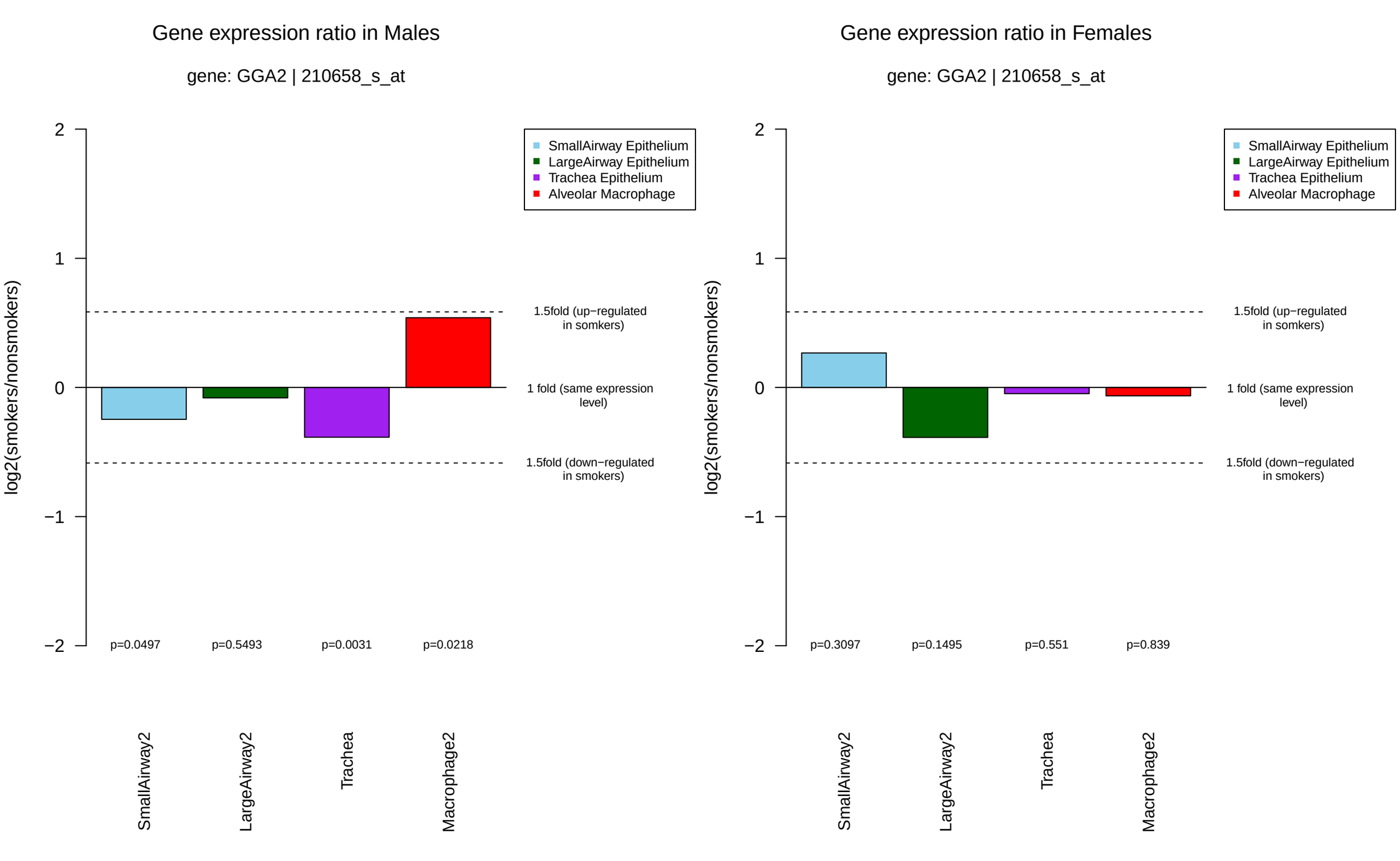
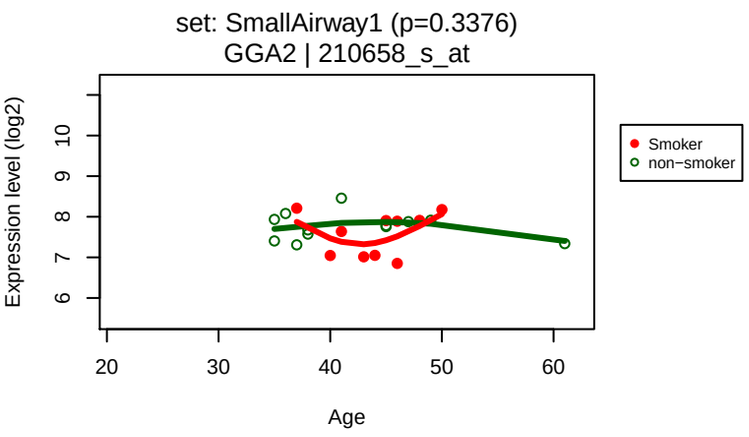
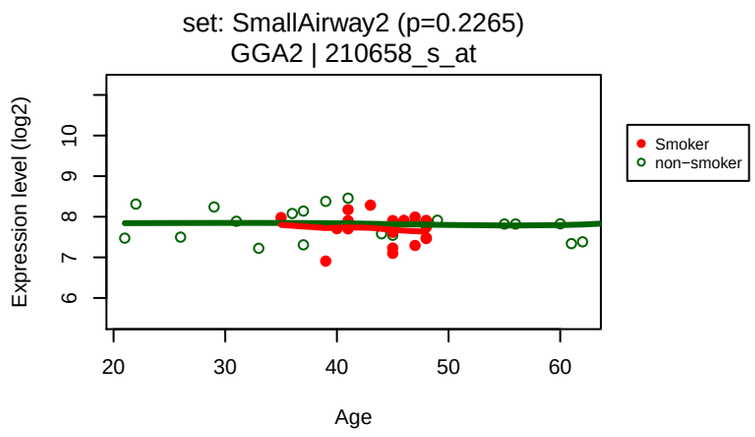
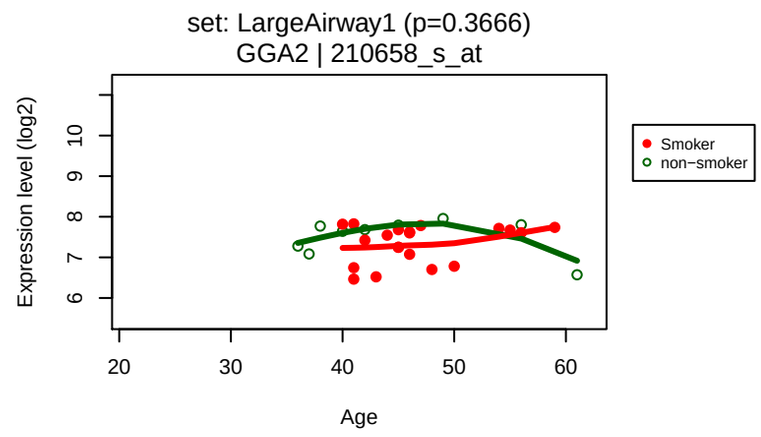

Gene expression of smokers and nonsmokers
gene: GGA2 | 210658_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3376
p=0.2265
p=0.1253
p=0.3666
p=0.0785
p=0.0026
p=0.1892
p=0.0778
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

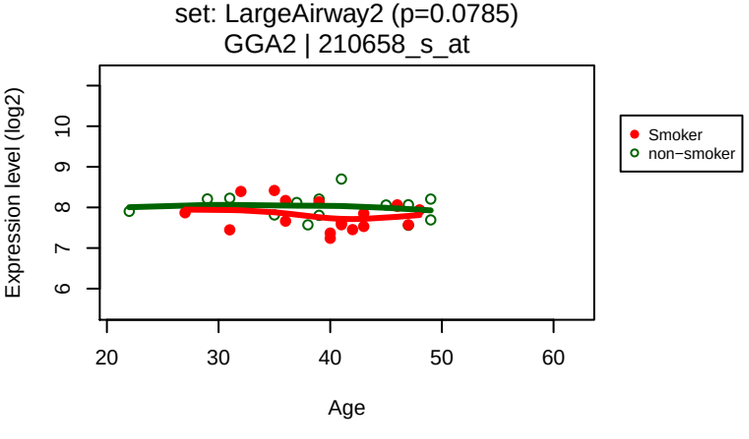

Gene expression ratio (smokers/nonsmokers)
gene: GGA2 | 210658_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3376
p=0.2265
p=0.1253
p=0.3666
p=0.0785
p=0.0026
p=0.1892
p=0.0778
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

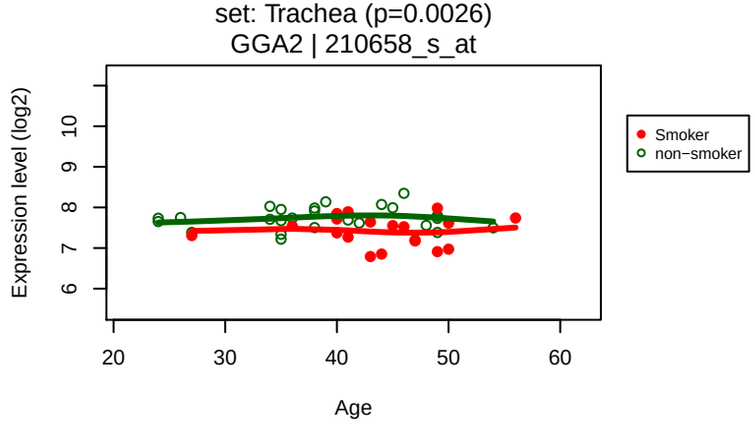

Gene expression ratio in Males
gene: GGA2 | 210658_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0497
p=0.5493
p=0.0031
p=0.0218
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GGA2 | 210658_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3097
p=0.1495
p=0.551
p=0.839
SmallAirway2
LargeAirway2
Trachea
Macrophage2

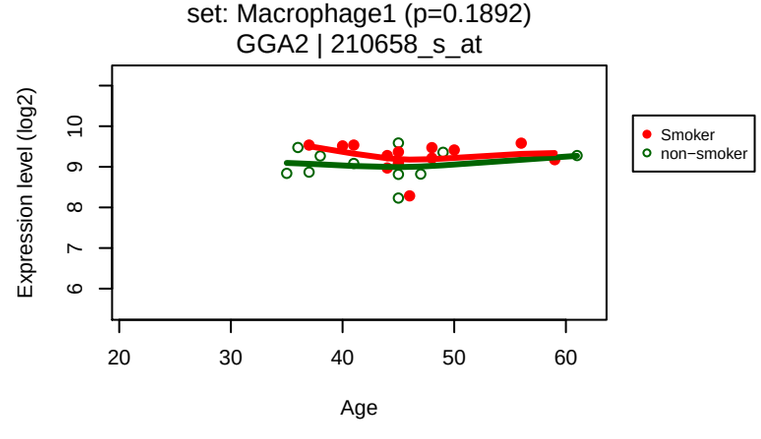

set: SmallAirway1 (p=0.3376)
GGA2 | 210658_s_at
Expression level (log2)
Age
Smoker
non−smoker

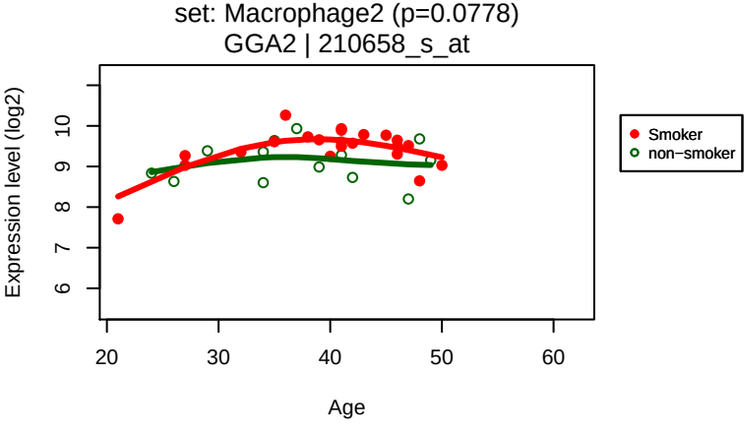

set: SmallAirway2 (p=0.2265)
GGA2 | 210658_s_at
Expression level (log2)
Age
Smoker
non−smoker

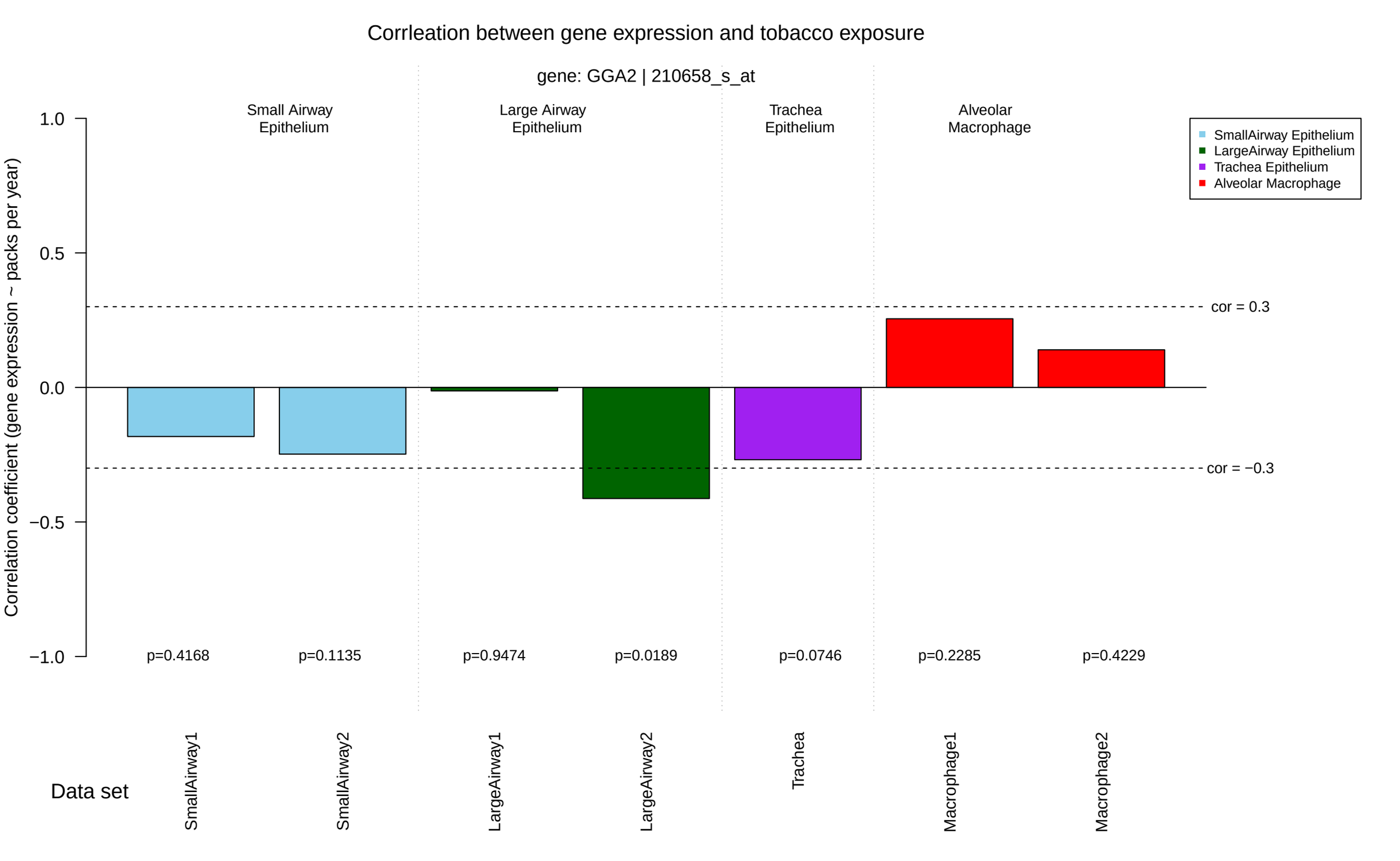

set: LargeAirway1 (p=0.3666)
GGA2 | 210658_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0785)
GGA2 | 210658_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0026)
GGA2 | 210658_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1892)
GGA2 | 210658_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0778)
GGA2 | 210658_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GGA2 | 210658_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4168
p=0.1135
p=0.9474
p=0.0189
p=0.0746
p=0.2285
p=0.4229
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2